

Time: Three Hours

Maximum Marks: 75

SECTION– A ($10 \times 2 = 20$ marks)**Answer any TEN questions**

1. Write down the various steps of planning.
2. Write any four characteristics of Statistics.
3. Write a few popular methods of business forecasting.
4. What are the types of diagrams you know?
5. Define pie diagram.
6. Write the five graphs of frequency distribution.
7. Find median for the following data.
6 9 21 5 7 2 0 32 9
8. Define mode.
9. Write down the Kelly's co-efficient of Skewness.
10. Write down the formula for Bowley's coefficient of Skewness.
11. Define positive correlation.
12. Define standard Error.

SECTION– B ($5 \times 5 = 25$ marks)**Answer any FIVE questions**

13. What are the essentials of good questionnaire?
14. Give a short note about merits of diagrammatic representation.
15. Calculate the mode.

Daily wages in Rs	50-60	60-70	70-80	80-90	90-100
No of Labours	40	62	75	100	65

16. Calculate weighted Geometric mean.

Commodity	A	B	C	D
Weight	1	6	3	2
Price (Rs)	5	17	30	42

17. The sum and the sum of the squares of 60 items are 1860 and 67100 respectively. Mode is 28.49. Find Pearson's coefficient of Skewness.
18. The first four moments of a distribution about the value 5 of a variable are 2, 20, 40 and 50. Find the arithmetic mean, variance, μ_3 and μ_4 .
19. Explain scatter diagram of correlation.

SECTION – C ($3 \times 10 = 30$ marks)

Answer any THREE questions

20. Explain about sampling and its types.

21. Draw a suitable diagram to represent the following data.

Item	Family A	Family B
Food	240	300
Clothing	160	300
Rent	80	200
Other Expenses	200	400

22. Find quartile deviation value.

X	351-500	501-650	651-800	801-950	951-1100
f	48	189	88	47	28

23. Calculate μ_1, μ_2, μ_3 and μ_4 .

Interval	0-9	10-19	20-29	30-39	40-49	50-59	60-69
Frequency	3	10	40	100	45	32	20

24. Calculate correlation coefficient from the following results.

$$N = 0, \quad \sum x = 140, \sum y = 150, \sum (x - 10)^2 = 100, \sum (y - 15)^2 = 215,$$

$$\sum (x - 10)(y - 15) = 60.$$